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Chapter 1

Introduction

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Chapter 1

Introduction

Classes of Computers and Early History of Computing

Learning Objectives (1.1)

1. Describe the following classes of computers:
 - Supercomputers
 - Microcomputers
 - Embedded systems
2. Describe two early computer architecture designs.

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What is a Computer?

- From supercomputer → server → PC → tablet → mobile phone → watch.
- All these devices contain some form of computational elements.
- No definitive way to classify computers. But we review three broad categories:

Supercomputer, microcomputers and embedded systems.



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Classes of computers Supercomputers

- Very large, powerful and expensive computers.
- High computational performance and can operate on large data sizes (for high precision calculations).
- Generally scalable by adding more processors.
- Applications - weather forecasting, simulation of complex physical systems and sub-atomic structures.

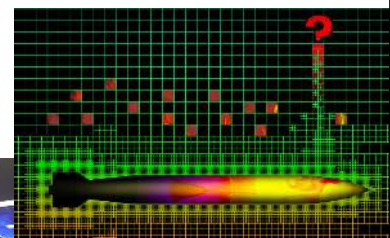
The Titan Supercomputer

at Oak Ridge National Laboratory, USA

Computational peak performance is around 17-27 petaFLOPS.

Titan consist of

- 18,000+ Nvidia Tesla K20 GPUs
- 700 terabytes of memory



Assess Health of Nuclear Bombs

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Classes of computers

Microcomputers

- Microcomputers contain a microprocessor as a processing unit and external memory and peripheral chip support.
- More powerful **workstations** are used as servers and the more common variety such as desktop **PC** and **notebooks** are for home-office computing applications.

High-end
ServerPersonal
Computer

Notebook

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Classes of computers

Embedded Systems

- Compact devices that usually employ a single-chip (**microcontroller**) containing the processing unit, memory and relevant peripheral support.
- They are called **embedded** systems as the presence of the microprocessor is non-obvious. Such devices are all around us.



Examples of embedded systems

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Early Days of the Digital Computer



- Major progress made during World War II (1940's)
- Computer research funded mainly by the War Department
- To solve problems related to ballistics

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Typical Ballistic Computation



Analog gear-based computer

- Knobs input numbers such as target speed and course, range to target, wind speed, wind direction, own speed, own course, etc. The outputs controlled the motors of the gun.

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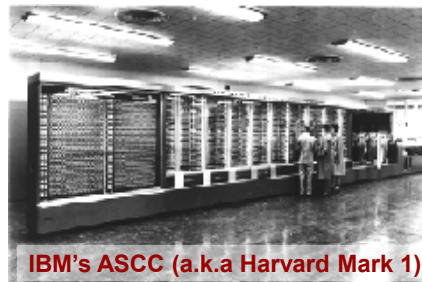
Harvard and Von Neumann

- Two major classes of computer architecture emerged.
- Harvard architecture**, named after Harvard series of relay calculators developed by Howard Aiken at Harvard Univ.

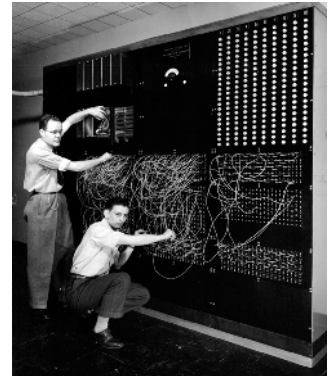


Howard Aiken

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IBM's ASCC (a.k.a Harvard Mark 1)

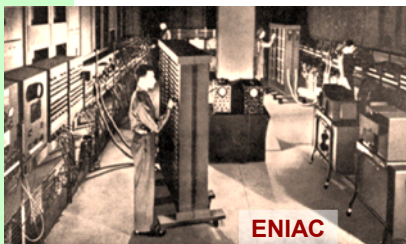


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Harvard and Von Neumann

- Two major classes of computer architecture emerged.
- Harvard architecture**, named after Harvard series of relay calculators developed by Howard Aiken at Harvard Univ.
- Von Neumann architecture**, developed by John Von Neumann at Princeton University. Influenced ENIAC's design.



ENIAC

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John von Neumann

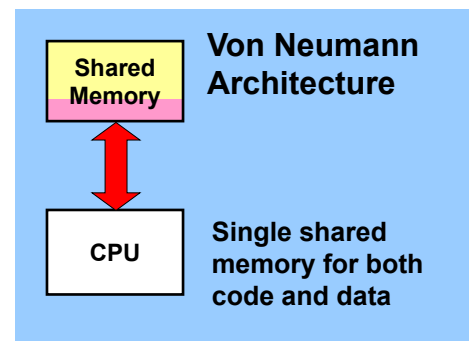
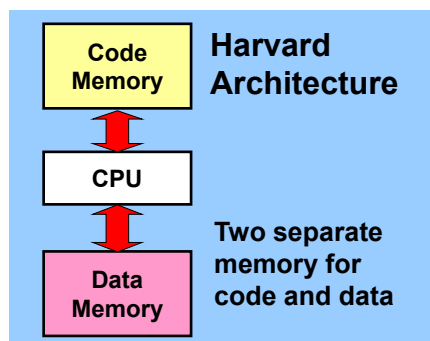


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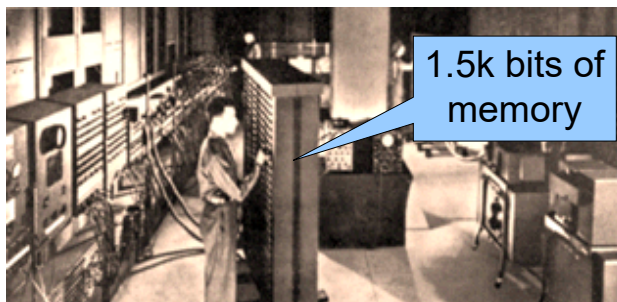


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ENIAC – the first digital computer



Specifications:

Weighed 30 tons, contained 19,000 vacuum tubes, 1,500 relays and consumed 200kW of power.

Electronic Numerical Integrator and Calculator

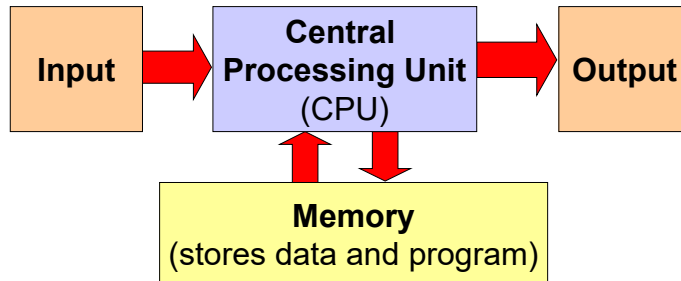
- In 1943, the US army funded Presper Eckert and John Mauchly at Univ. of Pennsylvania to build ENIAC, based on **von Neumann's architecture**.

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The von Neumann Architecture



- Many modern day computers are still based on von Neumann's design, which consist of:
 - Central Processing Unit (CPU)
 - Memory
 - Input and Output

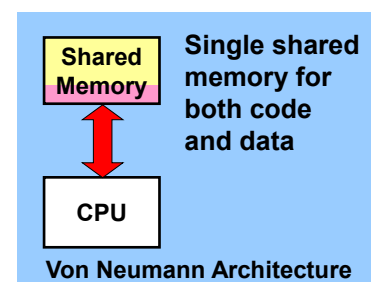
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Summary

- Computers can be classified in many ways, e.g. by function, size, general design, etc.
- We looked at three classes, namely supercomputers, microcomputers and embedded systems.
- Two early rivals in computer architecture designs, the **Harvard** and **von Neumann** architectures.
- In part, due to the high cost of memory in the early days of computing, the shared memory design of the von Neumann design became the preferred architecture.



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Chapter 1

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Basic Components of a Microcomputer

Learning Objectives (1.2)

1. Describe the basic components of a microcomputer.
2. Describe the purpose of the CPU clock and reset circuitry.

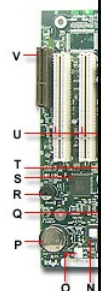
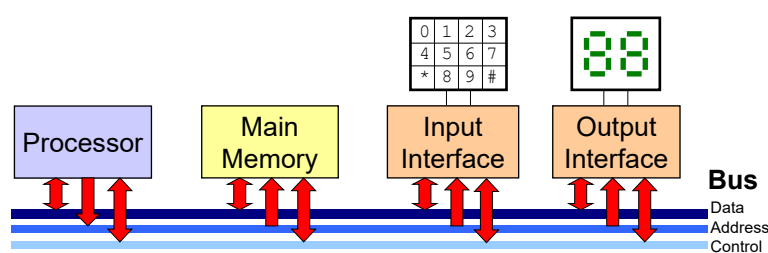
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Components of a Microcomputer



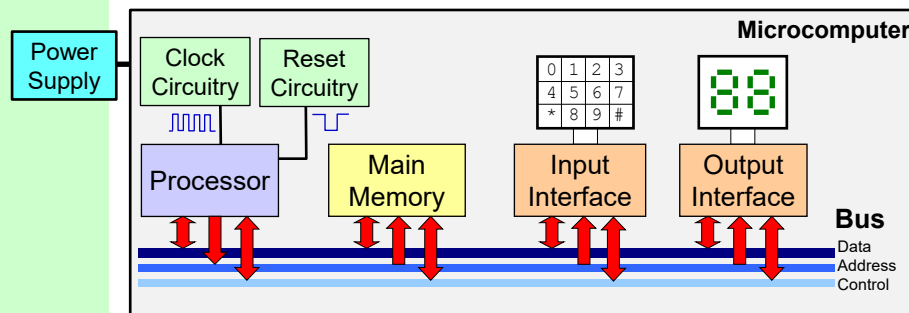
- Consist of three main components: **processor**, main **memory** and **I/O** interfaces.
- They are interconnected by a **bus** structure, which consist of a collection of wires through which binary information can be transferred in parallel.

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Components of a Microcomputer



- Consist of three main components: **processor**, main **memory** and **I/O** interfaces.
- They are interconnected by a **bus** structure, which consist of a collection of wires through which binary information can be transferred in parallel.
- Other important components include the **power** supply, CPU **clock** and **reset** circuitries.

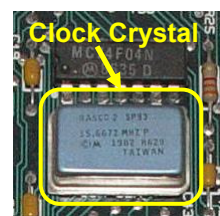
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Clock

- Most computers are **synchronous** and are driven by a master or system **clock**.
- The **speed performance** of the computer is governed by the frequency of the clock.
- The CPU requires a fixed number of clock ticks (**cycles**) to execute each instruction.
- Many **different clock** frequencies are derived from the one master clock.
- Operation closer to the CPU core (e.g. registers and arithmetic & logic units) are **clocked faster** and those involving external components (e.g. memory or peripheral access) are **clocked slower**.



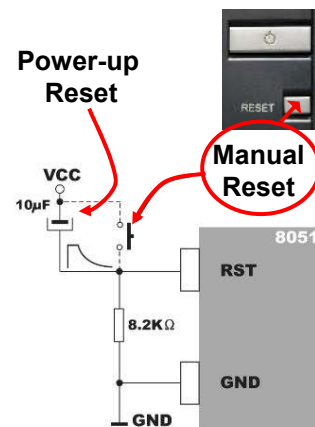
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Reset Circuitry

- The CPU is put into a known state on power up. The **reset circuitry** provides an external signal that asserts the Reset pin when power is applied.
- An active-low signal on the reset pin for a **substantial duration** (several clock cycles) is required to reset the CPU.
- Most computer system provide an additional **manual reset** button to reset the CPU without switching off the power.
- On reset, the CPU is put into a known initial state where the boot-up code can then execute.



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Summary

- The basic components within a computer consist of the CPU, memory and I/O interfaces.
- The **memory** is a very critical component in a computer as it stores both data and instructions.
- The access speed of the memory usually determines the performance of the computer.
- A fast processor with a fast clock that is coupled with slow memory will still execute instructions slowly.
- Understanding how data and instructions are organised in memory can help programmers write more **efficient programs**.

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Chapter 1

Introduction

Desktop PC and Tablet PC Examples

Learning Objectives (1.3)

1. Describe the hardware composition of a desktop PC.
2. Describe the hardware composition of a tablet computer.

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Computer Hardware Decomposition

- What are the major components within the typical computers that we use?



**Desktop Personal
Computer**



**Tablet
Computer**

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Inside a Desktop Personal Computer

- Major components of a desktop PC



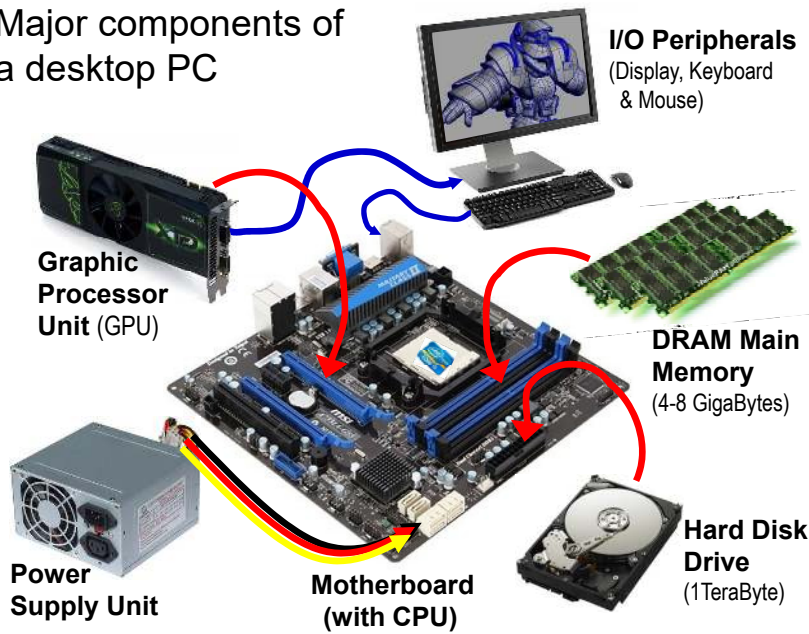
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Inside a Desktop Personal Computer

- Major components of a desktop PC



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Inside a Tablet Computer

- Major components of the iPad2



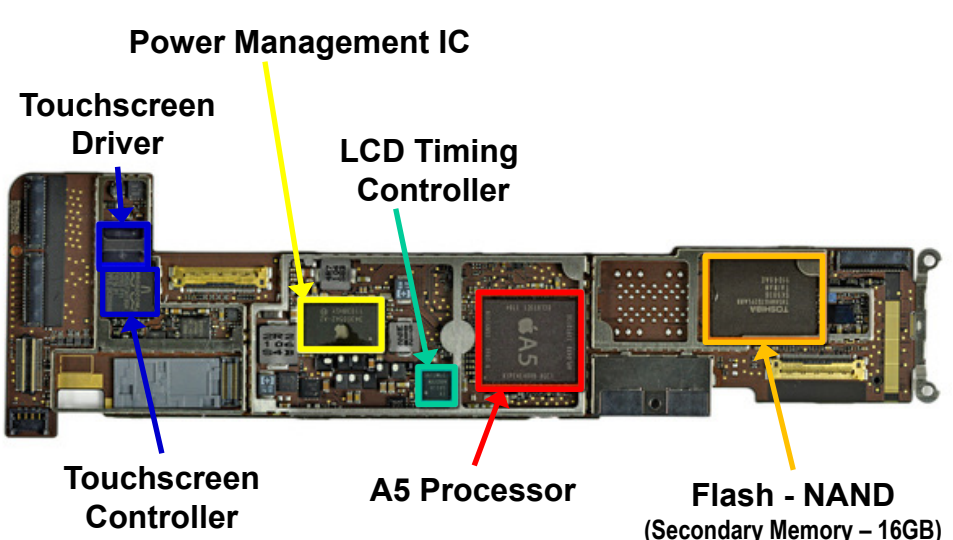
Source: http://www.appleinsider.com/articles/11/03/11/live_tear-down_of_apples_ipad_2_currently_underway.html

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Inside the iPad 2



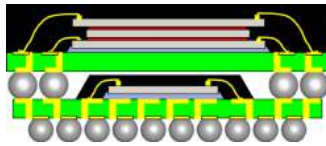
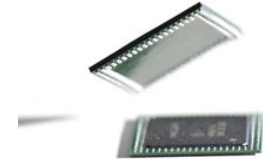
Source: http://www.appleinsider.com/articles/11/03/11/live_tear-down_of_apples_ipad_2_currently_underway.html

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Apple A5 Processor

- The A5 is a **package on package (PoP) system-on-a-chip (SoC)** that was designed by Apple and made by Samsung.



Package on package



Source: http://www.appleinsider.com/articles/11/03/15/x_ray_of_apples_a5_cpu_in_ipad_2_confirms_manufacturing_by_samsung.html

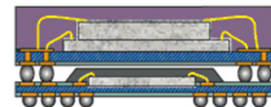
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Benefit of PoP Packaging

- Package on package (PoP) is an IC packaging technique that vertically stacks and interconnect separate packages (e.g. CPU and memory) via ball grid array (BGA) connections.
- Some benefits of PoP packaging:
 - **Save space** on motherboard - reduce size of product.
 - **Minimize track length** between CPU and memory - faster signal propagation and reduced electrical noise.
 - Memory units can be **tested separately** before combining with CPU units - improve manufacturing yield and supports multiple memory suppliers.
 - **Different-sized memory** can be coupled with CPU based on user requirements - simplifies inventory control.



Try: Google Search "Benefits of Package on Package"

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A5 Processor (System-on-a Chip)

- The A5 processor is with its built in I/O interfaces and support is considered a system-on-a-chip (SoC).
- A dual-core **ARM Cortex-A9** CPU with 4.5MB cache memory.
- 1GHz CPU clock, can be dynamically reduced to save battery life.
- 512MB low-power DDR SDRAM (@533MHz).
- Dual core PowerVX SGX543MP2 GPU to speed up graphics.



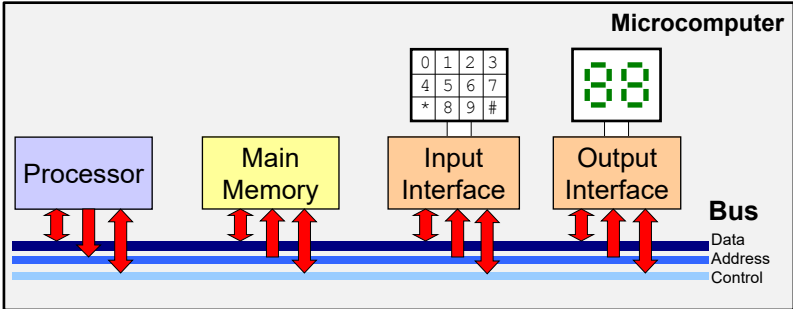
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Summary

- Whether a desktop or tablet PC, the basic components of a computer remains the same.
- These basic components are essentially the **CPU, memory** and the various **I/O interfaces** that permit peripherals to be connected to the computer.



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